

IE 442
MRP Implementation Report

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1- Data Used

The Chair (item_id = "001") is the finished product that requires several components for its assembly. Currently, there are 3 chairs in stock, and it takes 2 periods to assemble one chair. The components needed for a chair include 4 Legs (item_id = "002"), 1 Seat (item_id = "004"), 1 Back (item_id = "005"), and 2 Armrests (item_id = "003"). Legs are produced in lots of 10 and currently have an inventory of 10, which is sufficient to assemble 2 more chairs. The seat is ordered in lots of 5, with an inventory of 10, and the backrest is also ordered in lots of 5 but has only 4 units in stock, making it the limiting factor for chair production. The armrests are produced in lots of 5, and with an inventory of 20, there are enough armrests for 10 additional chairs. Based on the current stock of components, the production of chairs is constrained by the availability of backrests, limiting production to 4 more chairs unless more backrests are procured.

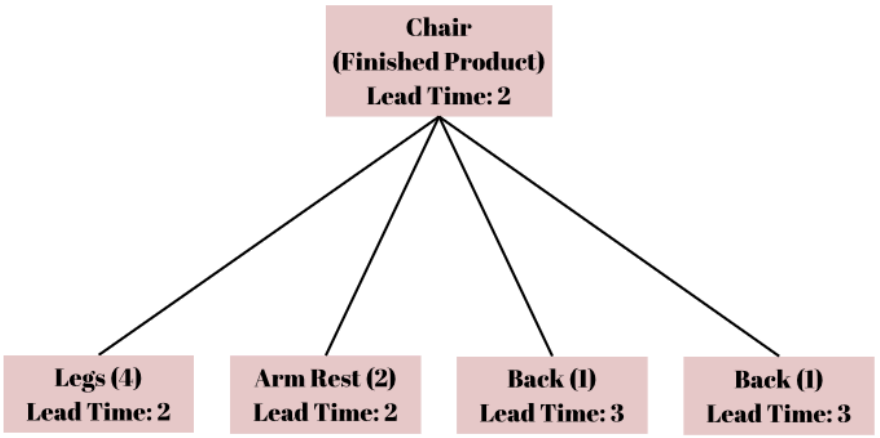


Figure 1: BOM of Chair

2- SQL Implementation

The SQL implementation provides the basis for an MRP system that integrates order processing, inventory management, and production planning. Included are the main tables: `item_period`-to track inventories and planned actions, `orders`-for customer orders, `required_items`-to calculate component requirements, and `bom`, the hierarchy of items and components. Procedures like the calculation of `GetRequiredItemsCount` recursively over the BOM, while `UpdateMRP` synchronizes inventory levels, requirements, and order plans across periods. A trigger, `order_come`, updates dependent tables in real time whenever a

new order arrives. In this way, complex planning is automated, which will make it possible to scale this for businesses with complex production requirements. Future enhancements could include indexing, better error handling, and improved concurrency management to further optimize performance and reliability.

Tables are as follows:

1. Customer

- customer_id (PK)
- customer_name

2. Order

- order_id (PK)
- customer_id (FK) → References Customer.customer_id
- period_number (FK) → References Periods.period_number
- item_id (FK) → References Item.item_id
- amount

3. Item

- item_id (PK)
- item_name
- lot_size
- lead_time
- inventory

4. BOM (Bill of Materials)

- bom_multiplier (PK)
- item_id (FK) → References Item.item_id
- component_id (FK) → References Item.item_id

5. Item_Period

- period_number (PK) → References Periods.period_number
- item_id (FK) → References Item.item_id

- gross_requirement
- projected_inventory
- planned_order_receipt
- planned_order_release

6. Periods

- period_number (PK)
- start_date
- end_date

7. Required_Items

- required_item (FK) → References Item.item_id
- required_amount
- due_period (FK) → References Periods.period_number

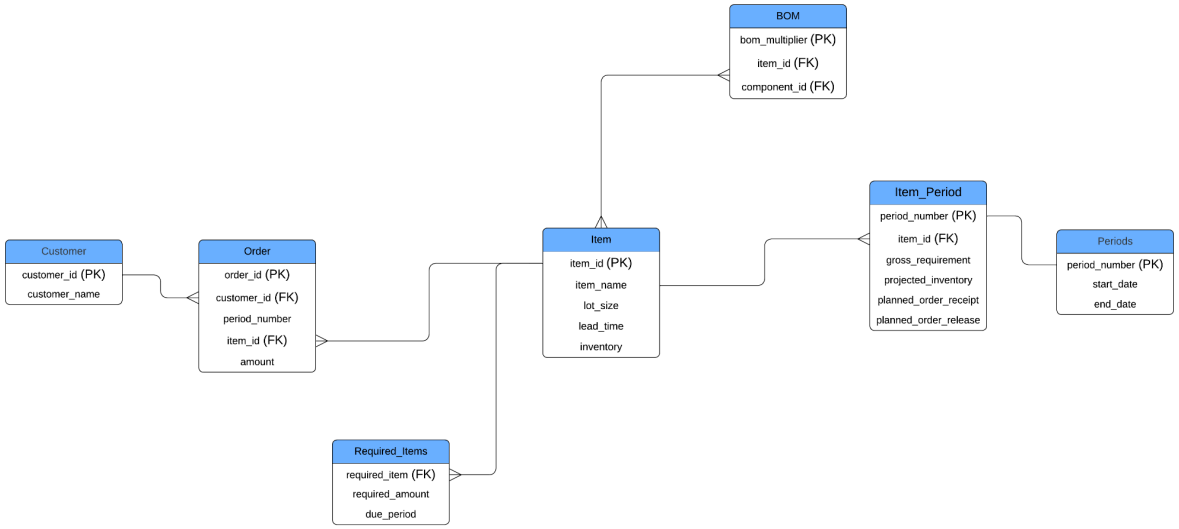


Figure 2: E-R Diagram

3- Python Implementation

The Python part serves to complement this SQL-based MRP system by facilitating operations with the database through an interface. The code connects to the database using `mysql.connector`, hence ensuring safe authentication and effective query execution. Its core functionalities include inserting customer orders, dynamically updating dependent tables with

SQL procedures, and fetching updated inventory and planning data from the `item_period` table. This will also enable real-time synchronization between the Python application and the database, further extending its usability and accessibility. It will also provide scalability for integrating with larger systems like ERP platforms or web-based interfaces to deliver a comprehensive solution for inventory and production management. Further development can be done on developing user-friendly dashboards, advanced analytics, and API-based integrations for broader applicability.

4- Improvements

It has been recognized that my proficiency in SQL is not at the desired level, which indicates a need for further development in this area. Although the table named Periods has been linked, issues were encountered in managing the inputs for the date fields. Consequently, this part of the implementation is currently not operational. It has become evident that this section requires additional work and improvements to ensure that date-related inputs function properly. Efforts will be directed toward addressing these challenges to enhance the system's performance and reliability.